

# Project Apex Race Report - Race Event

## Executive Summary

The race was decisively led by the Porsche 992 platforms, showcasing superior raw pace and effective tire management. The 992 P/A variant, notably Car #295 (Loftus Motorsports) and #298 (BR Racing), consistently delivered the fastest laps and maintained the best average green flag pace, with #295 setting the overall fastest lap at 1:27.896 and holding an average pace of 1:30.531. The 992 P/A class also demonstrated excellent driver consistency, reflected in a minimal 0.394-second gap between their fastest and slowest drivers. Competition Motorsports' Car #243 further highlighted this class's strength with the lowest tire degradation among all entries. The mid-field was primarily occupied by the Porsche Cayman M and P/A entries. Despite an inherent pace deficit of 3-6 seconds per lap compared to the 992s, several Cayman teams exhibited commendable driver consistency. Car #49 (Tatum Racing) achieved the best overall pace consistency (STDEV 0.547) despite navigating the most traffic, indicating strong driver execution. Cars #70, #73, and #27 also showed strong consistency, mitigating some of their raw speed disadvantage. Lagging performance was evident in specific entries across both platforms. Car #249 (Competition Motorsports, 992 M) was the slowest among the 992s in average pace (1:31.964) and showed higher inconsistency (STDEV 0.88). Car #204 (Paradaime Engineering, 992 P/A) suffered from the highest tire degradation, losing over 5 seconds in the final 5 laps due to wear, coupled with poor pace consistency (STDEV 2.04). In the Cayman class, Car #43 (CHR) was the slowest overall (1:37.293 average pace) with high inconsistency (STDEV 2.428), and Car #44 (Tatum Racing) exhibited extreme pace inconsistency (STDEV 9.361). The biggest strategic differentiator in this race was the intrinsic performance advantage of the Porsche 992 platform, synergized with superior tire management and consistent driver performance within the leading teams.

## Tactical Insights

- {'team\_type': 'leading', 'recommendation': 'Leverage the exceptional tire management of Car #243 (Competition Motorsports) to explore aggressive race strategies, such as pushing harder in the early laps or extending stints in multi-stint races.', 'justification': 'Car #243 demonstrated the lowest tire degradation with a 'deg\_coeff\_a' of 0.004094 and a predicted loss of only 0.567 seconds over the final 5 laps. This superior tire life provides a significant strategic advantage, allowing for higher average pace over a full stint or reduced tire changes in endurance formats.'}

- {'team\_type': 'mid-field', 'recommendation': "Capitalize on Car #49's (Tatum Racing) driver consistency and traffic management skills by optimizing car setup for consistent performance rather than peak raw speed. Focus on maintaining clean air and minimizing time loss in traffic, potentially through strategic positioning on track.", 'justification': "Car #49's driver, Greg Herback, exhibited the best race pace consistency (STDEV 0.547) despite encountering the most out-of-class traffic (9 laps). This indicates strong driver execution and adaptability, which can be maximized even with an inherent pace deficit to the leading class."}
- {'team\_type': 'lagging', 'recommendation': 'Conduct an immediate deep dive into tire management practices and car setup for Car #204 (Paradaime Engineering). Analyze driver inputs, suspension settings, and aerodynamic balance to mitigate the severe tire degradation observed.', 'justification': "Car #204 displayed the highest tire degradation with a 'deg\_coeff\_a' of 0.023964, resulting in a significant 5.401 seconds predicted loss over the final 5 laps. This, combined with a high race pace consistency STDEV (2.04), indicates a fundamental issue with how the car manages its tires or how the driver is utilizing them, critically limiting overall performance."}